Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*AeroHF – a new substrate for high-frequency application with zero loss*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (..), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), **Next-gen Communication (√),** Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Electrical and Electronic Components	Composite materials	Ceramic Materials	Ceramic Manufacturing Process Technologies
Concept of Proposed Technology	Aerogels represent a unique class of materials characterized by an open, three-dimensional network in which the liquid phase has been replaced by gas. Since their first introduction to the scientific community, aerogels have attracted sustained and growing research interest, particularly aimed at tailoring their chemical compositions and microstructures to fully exploit their exceptional physical properties, such as ultralow relative densities (< 0.5 g cm^{-3}), large specific surface areas (commonly in the range of 100–1000 m^2 g^{-1}), and remarkably low thermal conductivities (< 0.05 W m^{-1} K^{-1} at room temperature). Such properties position them as highly promising candidates for a wide spectrum of advanced applications, including energy storage systems, thermal insulation, sensing, and purification processes, etc.			

A key development of this research work is to advance process technologies to fabricate wafers that can be used and further processed in the electronic industry.

At Fraunhofer IZM glass and silicon wafers can be processed with laser drilling, plating and photolithographic etching to generate metallization structures based on Wafer-Level process technology, but not for ceramics. This technology is used for packaging of electronic components to have smallest structures for highest frequencies. A key development of this research activity is to develop process technologies for metallization, laser drilling, photolithographic processes for the new AeroHF substrate. Fraunhofer IZM will characterize the high-frequency properties of the materials to demonstrate the usability for telecommunication and mobile applications. Fraunhofer has the laboratory capability to characterize metallized structures up to 500GHz.

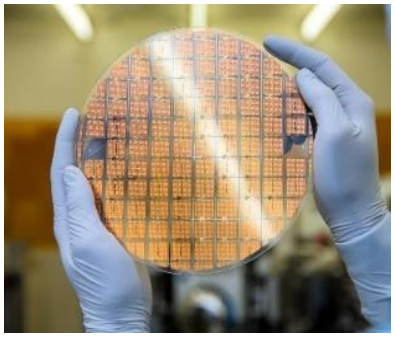


Figure 1: glass wafer with metallization structures

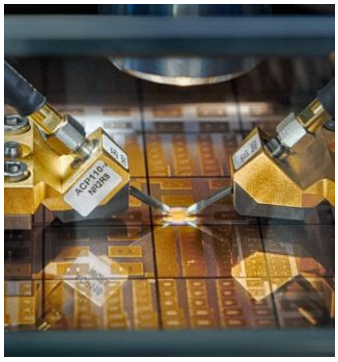


Figure 2: material characterization up to 110GHz on a processed wafer

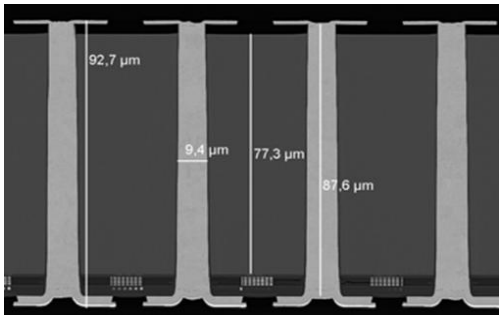


Figure 3: TGV: Through Glass Vias

Rationale for Joint R&D Project

Due to the lack of experience of ceramic processing at Fraunhofer IZM, an industrial partner in Koera with extensive knowledge in ceramic wafer processing, thin film processing, via drilling, plating and etching is required.

The proposed AeroHF is a PDC (“polymer-derived ceramic”) aerogel represents a unique dielectric material, combining low permittivity, minimal dielectric loss, and high thermal stability, that is

extremely beneficial for high frequency applications in electronics, such as 5G / 6G mobile communication, high-speed Ethernet 224Gbps for fiber-optical backbones and server farms, automotive radar (79GHz) and satellite communication. The very high pore volume (>90 vol%) , i.e., the materials basically holds air, brings the dielectric constant close to 1 and the dielectric loss close to 0, which is not feasible for any other material today. This is beneficial for high speed digital communications, were loss limits the speed of the data rate for chip-to-chip interconnects on substrates and interposers. For mobile communication the efficiency of antennas suffers due to the loss of the carrier substrate. New concepts such as A-i-P, Antenna-in-Package has been shown successful for future 6G mobile networks at 140GHz. Although the used mold or substrate material is an established material for chip packaging and is available for high-frequency applications, its Dk>1 reduces the radiation efficiency of the antenna and thus requires higher power consumption of the mobile device.

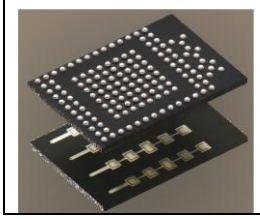


Figure 4: 140GHz A-i-P chip module for 6G manufactured at Fraunhofer IZM

In a long term the Korean ceramic PCB manufacturer shall be able to process AeroHF to provide similar chip package as shown above but with much better radiation efficiency for mobile communication or much lower signal degradation for digital communication.

Key Words

1. ceramic material development
2. ultra low loss, low Dk substrate material
3. high frequency applications, Ethernet 224Gbps, 5G / 6G mobile
4. new wafer level processes for ceramics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (1)	End of the Project	Level (4)

R&D Objectives

Polymer-derived ceramics (PDCs) are a class of materials obtained via preceramic polymer pyrolysis. Using the PDC route, metal silicates of the (Si-O-C-N)–M family can be produced, where M denotes a functional metal, through low-temperature, cost-efficient processing. This approach allows the formation of M-silicate PDC aerogels with low permittivity, minimal dielectric loss, high thermal stability, and tunable electrical conductivity suitable for high-frequency applications.

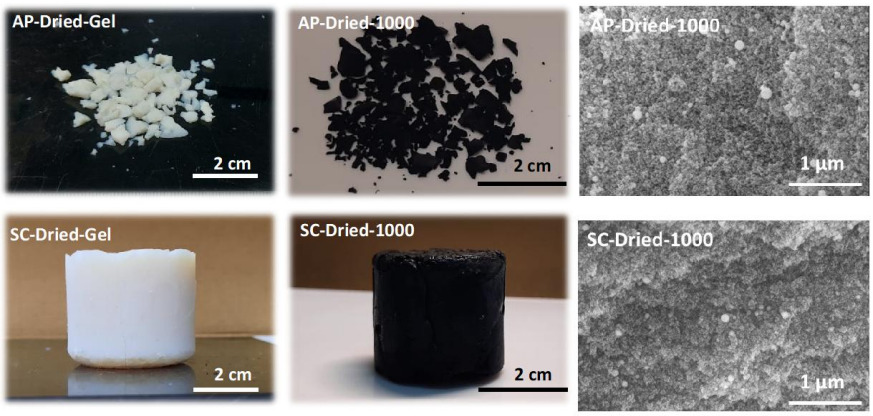


Figure 5: Aerogels formed via PDC route, top images are from chunks and bottom from monoliths.

The manufacturing of AeroHF as wafer substrate is challenging. Since aerogels are mainly `fragile` materials. Their processing is challenging, no processes for metallization, neither for VIA laser drilling, plating, etching are known. These processes need to be developed and investigated. However, aerogels produced via the PDC route exhibit notable mechanical strength, with compressive strength reaching ~3 MPa at ~90 vol% porosity. Their fabrication is considerably easier than that of conventional silica aerogels, as neither supercritical drying nor surface functionalization is required. This renders PDC aerogels particularly suitable for large-scale production, offering enhanced processing speed and cost-effectiveness.

Fraunhofer IZM has extensive experience in Wafer level CSP / SiP: Cu redistribution, organic/inorganic dielectrics, TGV Through Glass Vias, TSV Through Silicon Vias, additive microstructuring, package singulation, wafer and die thinning, fan-in and fan-out wafer level packaging, reliability investigation and understands the requirements for wafer processing very well.

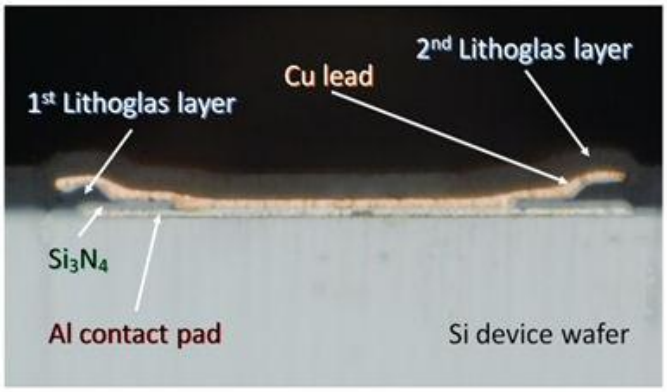


Figure 6: multilayer interconnects based on thin Cu and glass

The characterization of materials with Dk ~ 1 and dielectric loss ~ 0 is extremely challenging. Fraunhofer IZM has several methodologies for unprocessed raw materials that requires quite thin wafers below 200µm. Other

methodologies require processed wafers with tight tolerances. Resonator measurements with dielectric loss tan∂ < 0.0002 has been shown for PTFE (Teflon) up to 140GHz but at Dk~2. Materials with lower Dk values were not yet available.

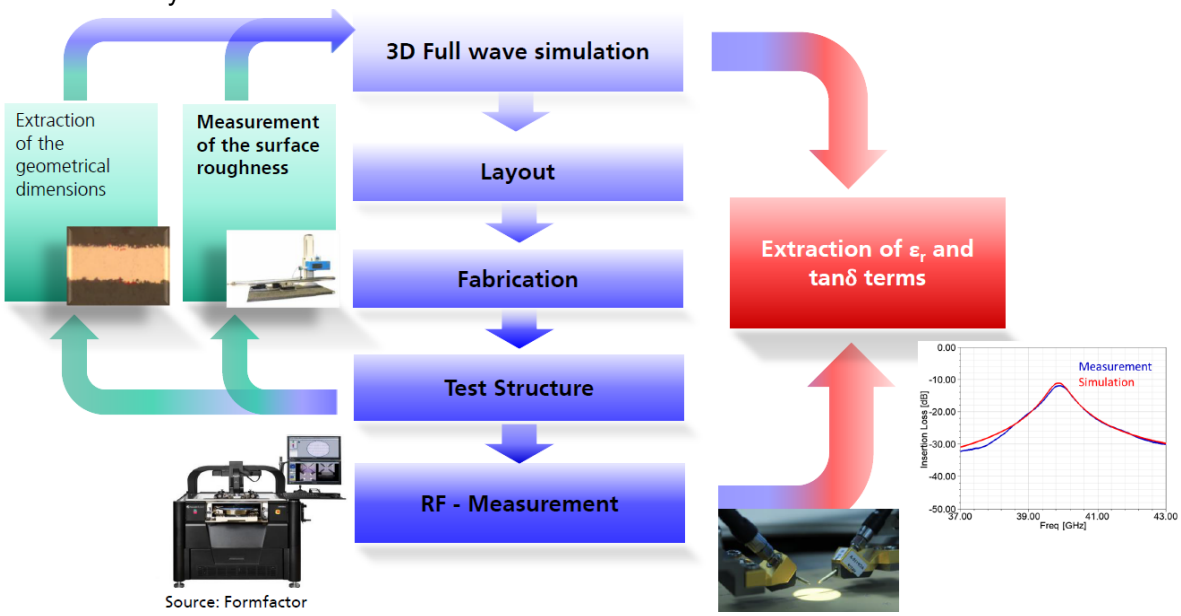
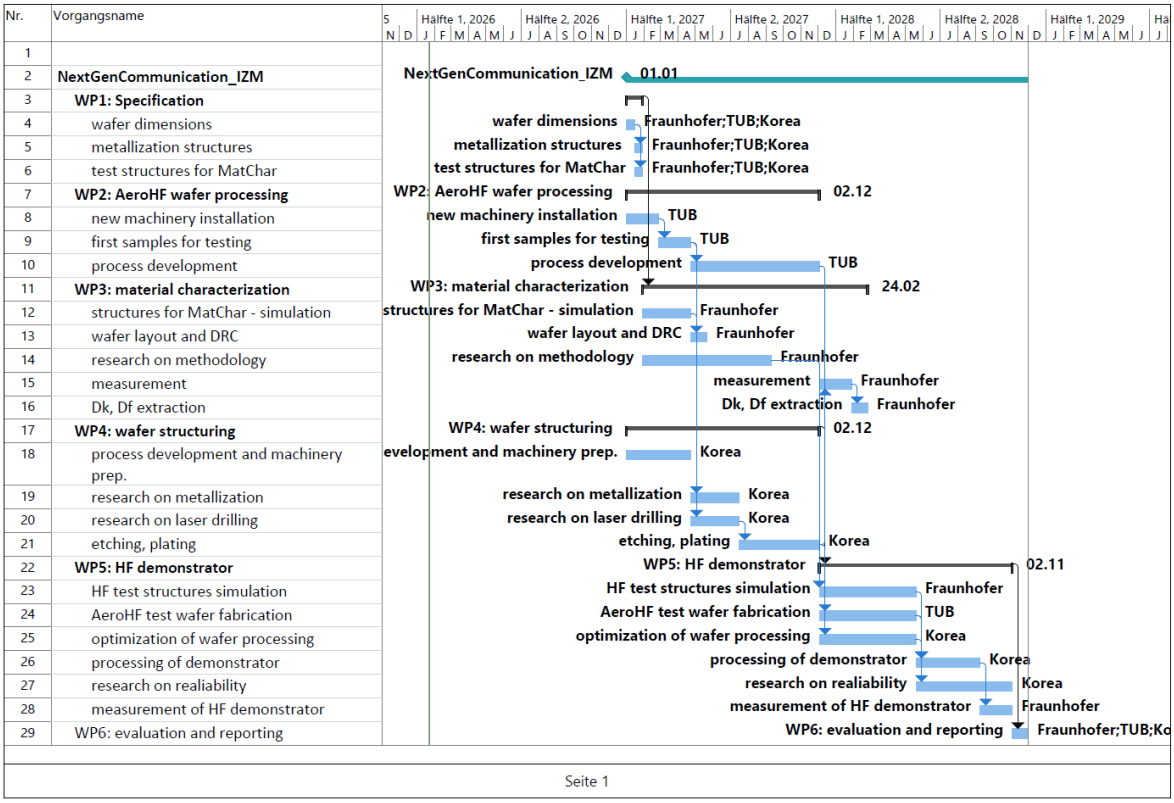


Figure 7: Extraction process of metallized substrates

R&D Target Specifications	Process development at TU Berlin, Chair of Advanced Ceramic Materials Total porosity > 80 vol%, specific surface area ≥ 100 m^2/g, at RT thermal conductivity below 0.05 W m^{-1} K^{-1}, Wafer diameter up to 4 inch Wafer thickness < 500µm Process development at Korean industry partner Metallization structure Lines/Spaces < 50µm Through-Ceramic-Vias < 200µm Methodology development at Fraunhofer IZM Extraction of dielectric parameters Dk close to 1 and Df close to 0 up to 110GHz or beyond High-frequency interconnects for frequencies > 110GHz

R&D Plan



WP1 specifies technical details such as minimum wafer size required for processing, metallization (lines and spaces) and via dimension to achieve high-frequency structures such as microstrip and coplanar waveguides

WP1 and WP3 TUB and Fraunhofer focus on material characterization, how to process suitable wafer slices and the development of material characterization methodology

WP1 and WP4 TUB and Korean industry partner focus on processing the wafers, investigating critical process steps, machinery optimization and identify process challenges

WP5 TUB, Fraunhofer and Korean industry partner develop a demonstrator to show high-frequency capability and finally testing the HF performance

WP6: assess technical achievements and future commercialization

MS1: material characterization finished, end of WP3

MS2: wafer processing accomplished, end of WP4

MS3: demonstrator analyzed, end of WP6

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TUB – Technsche Universitaet Berlin, Chair of Advanced Ceramic Materials Process development of AeroHFwafers Fraunhofer IZM Proves development of vias, metallization, structuring Development of material characterization methodology at high frequencies for ultra low Dk, Df values Measures the test structures up to 110GHz and beyond with wafer probes
Expected R&R of Korean SME Partner	A transfer of knowledge from Chair of Advanced Ceramic Materials, TUB and Fraunhofer IZM to Korean SME how to process AeroHF wafers is targeted. This will only be possible after intellectual property rights (patent application) have been secured for AeroHF production and the process development of AeroHF wafers for the Chair of Ceramic Materials (TUB). The Korean SME will develop its own IP on further processing structures on the AeroHFwafer for High-Frequency applications. A partner such as OTFT Technology (www.otftechnology.com) is preferred with extensive knowledge in ceramic wafer processing, thin film processing, via drilling, plating and etching. A proper clean room facility for photolithography is required. The Korean SME will validate the developed processes and refine them to a commercialized level.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc. Chair of Advanced Ceramic Materials, TU Berlin will generate knowledge about procedures and methods to manufacture AeroHF as wafers for the electronic industry. Fraunhofer IZM will generate knowledge how to extract dielectric material parameters of a processed AeroHF wafer and provides interconnect designs for high-frequency applications. In the cooperation with a Korean SME a roadmap from research to commercialization of AeroGel as a new substrate material for high frequency applications will impact the global electronic industry by providing a new substrate with extraordinary behavior. It allows Korean semiconductor industry to improve its global position in the electronic industry by providing electronic components with better radiation efficiency and thus lower power consumption and higher data rates for digital semiconductors in communication and memory applications.

6. References

(1) ACS Applied Materials & Interfaces, 2020, Maged F Bekheet, Aleksander Gurlo, “Polymer-derived SiOC integrated with a graphene aerogel as a highly stable Li-ion battery anode”

(2) "Freeze Casting: From Low-Dimensional Building Blocks to Aligned Porous Structures-A Review of Novel Materials, Methods, and Applications" (2020), Advanced Materials

(3) IEEE, 2024, Prof. Ndip, Dr. Koeszegi, Jens Schneider, „Temperature dependent Dielectric Characterization of Low Loss Thin Film Polymer up to sub-THz bands”

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization